We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**
BUTYL ACRYLATE

Use: Monomer.

Uses advised against: All consumer uses are strongly advised against., Use of substance in adhesives (professional), Use of substance in coatings (professional), Use of substance in inks and toners (professional)
Recommended use: for industrial use only
Not recommended use: cosmetics, Pharmaceutical

Manufacturer/supplier:
BASF Japan Ltd.
OVOL Nihonbashi Building 3F
3-4-4 Nihonbashi Muromachi, Chuo-ku
Tokyo, 103-0022, JAPAN
Telephone: +81-3-5290-3000
Telefax number: +81-3-5290-3333

Emergency information:
Telephone: 03-6634-2245
+49 180 2273-112 (International emergency number)

## 2. Hazard identification

Classification of the substance and mixture:
Flammable liquids: Cat.3
Acute toxicity: Cat.4 (Inhalation - vapour)
Acute toxicity: Cat.5 (oral)
Acute toxicity: Cat.5 (dermal)
Skin irritation: Cat.2
Eye irritation: Cat.2A

Skin sensitization: Cat.1B
Specific target organ toxicity — single exposure: Cat.3 (irritating to respiratory system)
Hazardous to the aquatic environment - acute: Cat.2
Hazardous to the aquatic environment - chronic: Cat.3

Label elements and precautionary statement:

Pictogram:

Signal Word:
Warning

Hazard Statement:

| | |
|---|---|
| H226 | Flammable liquid and vapour. |
| H319 | Causes serious eye irritation. |
| H315 | Causes skin irritation. |
| H332 | Harmful if inhaled. |
| H317 | May cause an allergic skin reaction. |
| H335 | May cause respiratory irritation. |
| H303 + H313 | May be harmful if swallowed or in contact with skin. |
| H412 | Harmful to aquatic life with long lasting effects. |
| H401 | Toxic to aquatic life. |

Precautionary Statements (Prevention):

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|---|---|
| P280 | Wear protective gloves and eye protection or face protection. |
| P271 | Use only outdoors or in a well-ventilated area. |
| P260 | Do not breathe mist or vapour. |
| P280 | Wear eye protection. |
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P273 | Avoid release to the environment. |
| P243 | Take action to prevent static discharges. |
| P241 | Use explosion-proof electrical, ventilating and lighting equipment. |
| P272 | Contaminated work clothing should not be allowed out of the workplace. |
| P264 | Wash contaminated body parts thoroughly after handling. |
| P242 | Use non-sparking tools. |
| P240 | Ground and bond container and receiving equipment. |

Precautionary Statements (Response):

| | |
|---|---|
| P312 | Call a POISON CENTER or physician if you feel unwell. |
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P303 + P361 + P353 | IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. |
| P333 + P313 | If skin irritation or rash occurs: Get medical attention. |
| P302 + P312 | IF ON SKIN: Call a POISON CENTER or a doctor/physician if you feel unwell. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |
| P332 + P313 | If skin irritation occurs: Get medical attention. |
| P337 + P313 | If eye irritation persists: Get medical attention. |
| P370 + P378 | In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction. |

Precautionary Statements (Storage):

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|---|---|
| P403 + P235 | Store in a well-ventilated place. Keep cool. |
| P233 | Keep container tightly closed. |
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/information on ingredients

Chemical nature

Substance nature: single product

n-butyl acrylate
CAS Number: 141-32-2
ENCS: (2)-989
ISHL: (2)-989

**Hazardous ingredients**

n-butyl acrylate

| | |
|---|---|
| Content (W/W): >= 99.5 % - <= 100 %<br>CAS Number: 141-32-2<br>ENCS: (2)-989<br>ISHL: (2)-989 | Flam. Liq.: Cat. 3<br>Acute Tox.: Cat. 4 (Inhalation - vapour)<br>Acute Tox.: Cat. 5 (oral)<br>Acute Tox.: Cat. 5 (dermal)<br>Skin Irrit.: Cat. 2<br>Eye Irrit.: Cat. 2A<br>Skin Sens.: Cat. 1B<br>STOT SE: Cat. 3 (irr. to respiratory syst.)<br>Aquatic Acute: Cat. 2<br>Aquatic Chronic: Cat. 3 |

## 4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
dry powder, water spray, carbon dioxide, foam

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Use extinguishing measures to suit surroundings.

Specific hazards:
Risk of violent self-polymerization if overheated in a container. Cool endangered containers with water-spray.

The product is combustible. See SDS section 7 - Handling and storage.

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Extend fire extinguishing measures to the surroundings. Fight fire from maximum distance. Vapours are heavier than air and may accumulate in low areas and travel a considerable distance up to the source of ignition.

Further information:
In case of a fire in the vicinity a restabilization system should be used if the temperature in the bulk storage-tank reaches 45°C. Evacuate area of all unnecessary personnel. In case of a fire in the vicinity evacuate all personnel in a greater area if the temperature in the bulk storage-tank reaches 60°C.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

Personal precautions:
Handle in accordance with good industrial hygiene and safety practice.

Avoid all sources of ignition: heat, sparks, open flame.Use antistatic tools.

Environmental precautions:
Discharge into the environment must be avoided.

Methods for cleaning up or taking up:
For large amounts: Pump off product.
Spills should be contained, solidified, and placed in suitable containers for disposal. Dispose of absorbed material in accordance with regulations. Ensure adequate ventilation. Suppress gases/vapours/mists with water spray jet. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Cleaning operations should be carried out only while wearing breathing apparatus. Pick up with suitable appliance and dispose of.

Additional information: High risk of slipping due to leakage/spillage of product.

Release of substance/product can cause fire or explosion. Shut off or stop source of leak. Shut off or stop released substance/product under safe conditions.

Pack in tightly closed containers for disposal.

## 7. Handling and Storage

Handling

The substance/ product may be handled only by appropriately trained personnel. Facility parts must be checked for polymer residues and cleaned on regular basis in order to avoid hazardous reactions.

Ensure thorough ventilation of stores and work areas. Encapsulation or exhaust ventilation required. When filling, transferring, or emptying of containers, adequate local exhaust ventilation is necessary. Vent waste air to atmosphere only through suitable separators. Check the condition of seals and connector screw threads.

The temperatures which must be avoided are to be considered. Protect against heat. Protect from direct sunlight. Protect contents from the effects of light. Do not open warm or swollen product containers. Remove persons to safety and alert fire brigade.

Ensure adequate inhibitor and dissolved oxygen level.

Avoid inhalation of dusts/mists/vapours. Avoid aerosol formation. Avoid all direct contact with the substance/product.

Protection against fire and explosion:
Avoid all sources of ignition: heat, sparks, open flame. Substance/product can form explosive mixture with air. Ground all transfer equipment properly to prevent electrostatic discharge. It is recommended that all conductive parts of the machinery are grounded. Explosion-proof equipment is not necessary when loading and processing of the product takes place at a minimum of 5 °C below the flash point.

Heated containers should be cooled to prevent polymerization. If exposed to fire, keep containers cool by spraying with water. Emergency cooling must be provided for the eventuality of a fire in the vicinity.

Storage

Further information on storage conditions: Prior to storage ensure that the transfer equipment used and the intended storage containers do not contain other substances/products. Before transfer to stock the identity of the product must be proved to be without doubt. The entrance to storage rooms is to be granted only to appropriately trained personnel.
The stabilizer is only effective in the presence of oxygen. Maintain contact with atmosphere containing 5 - 21% oxygen. Never use tanks with inert-gas installation for storage.
Risk of polymerization. Protect against heat. Protect from direct sunlight. Avoid UV-light and other radiation with high energy. Protect against contamination.
In case of bulk storage, the storage-tanks should at least be equipped with two high temperature alert devices.
Even if the product is stored and handled as prescribed/indicated it should be used up within the indicated duration of storage.

Storage stability:
Storage temperature: < 35 °C
Storage duration: 12 Months
The stated storage temperature should be noted.
Avoid prolonged storage.
This product should be processed as soon as possible.
Ensure adequate inhibitor and dissolved oxygen level.

Do not store with less than 10 % headspace above liquid.
Storage stability is based upon ambient temperatures and conditions described.
It is recommended to keep a safe distance of +2 degrees above the crystallization range.
The product is stabilized, the shelf life should be noted.
Storage temperature: 45 °C
A restabilization system should be used if the temperature in the bulk storage-tank reaches the indicated value.
Storage temperature: 60 °C
All personnel in a greater area should be evacuated if the temperature in the bulk storage-tank reaches the indicated value.

## 8. Exposure controls and personal protection

Components with occupational exposure limits

n-butyl acrylate, 141-32-2;
TWA value 2 ppm (ACGIHTLV)
(OEL (JP))
Included in the regulation, but with no data values - See the regulation for further details

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
butyl rubber (butyl) - 0.7 mm coating thickness
nitrile rubber (NBR) - 0.4 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
Avoid contact with the skin, eyes and clothing. Avoid inhalation of vapour. Wearing of closed work clothing is required additionally to the stated personal protection equipment.

## 9. Physical and Chemical Properties

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|---|---|---|
| Form: | liquid | |
| Colour: | colourless | |
| Odour: | acrylic-like | |
| Odour threshold: | not determined | |
| pH value: | not applicable | |
| Melting point: | -64.6 °C<br>Literature data.<br>The substance / product does not decompose. | |
| Boiling point: | 147 °C<br>(1,013 hPa) | |
| Flash point: | 38 °C | (DIN 51755, closed cup) |
| Evaporation rate: | Value can be approximated from Henry's Law Constant or vapor pressure. | |
| Flammability (solid/gas): | Flammable liquid and vapour. | (derived from flash point) |
| Lower explosion limit: | For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point. | |
| Upper explosion limit: | For liquids not relevant for classification and labelling. | |
| Ignition temperature: | 275 °C | (DIN 51794) |
| Thermal decomposition: | No decomposition if stored and handled as prescribed/indicated. | |
| Self ignition: | Based on its structural properties the product is not classified as self-igniting. | Test type: Spontaneous self-ignition at room-temperature. |
| Self heating ability: | not applicable, the product is a liquid | |
| SADT: | Not a substance/mixture liable to self-decomposition according to GHS. | |
| Explosion hazard: | Based on the chemical structure there is no indication of explosive properties. | |
| Fire promoting properties: | Based on its structural properties the product is not classified as oxidizing. | |
| Vapour pressure: | 5 hPa<br>(22.2 °C) | |

| | | |
|---|---|---|
| Density: | 0.899 g/cm3<br>(20 °C)<br>0.8689 g/cm3<br>(50 °C)<br>0.8639 g/cm3<br>(55 °C) | (calculated) |
| Relative density: | 0.9<br>(20 °C) | |
| Relative vapour density (air): | 4.41<br>(20 °C)<br>Heavier than air. | (calculated) |
| Solubility in water: | 1.7 g/l<br>(20 °C) | |
| Solubility (qualitative) solvent(s): | organic solvents<br>miscible | |
| Partitioning coefficient n-octanol/water (log Pow): | 2.38<br>(25 °C) | (measured) |
| Adsorption/water - soil: | KOC: 88.4; log KOC: 1.95<br>Adsorption to solid soil phase is not expected. | (OECD Guideline 106) |
| Surface tension: | Based on chemical structure, surface activity is not to be expected. | |
| Viscosity, dynamic: | 0.88 mPa.s<br>(20 °C)<br>0.66 mPa.s<br>(40 °C) | |
| Viscosity, kinematic: | (20 °C)<br>not determined | |
| Molar mass: | 128.17 g/mol | |

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

## 10. Stability and Reactivity

Conditions to avoid:
Avoid heat. Avoid oxygen content above the product of less than 5 %. Avoid UV-light and other radiation with high energy. Avoid direct sunlight. Avoid prolonged storage. Avoid inhibitor loss. Avoid excessive temperatures. Avoid all sources of ignition: heat, sparks, open flame. Avoid freezing. Avoid moisture.

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.

Substances to avoid:
radical formers, free radical initiators, peroxides, mercaptans, nitro-compounds, perborates, azides, ether, ketones, aldehydes, amines, nitrates, nitrites, oxidizing agents, reducing agents, strong bases, alkaline reactive substances, acid anhydrides, acid chlorides, concentrated mineral acids, metal salts
Inert gas

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:
Explosion and fire hazard exists under confined conditions. Ignitable air mixtures can form when the product is heated above the flash point and/or when sprayed or atomized. Formation of explosive gas/air mixtures.
Polymerization coupled with heat formation.
Risk of spontaneous polymerization by oxygen depletion of the liquid phase. Risk of spontaneous polymerization when heated or in the presence of UV radiation. Risk of spontaneous and violent self-polymerization if inhibitor is lost or product is exposed to excessive heat. Polymerization produces gases which may burst closed or confined containers. Reactions may cause ignition.
Risk of spontaneous polymerization in the presence of starters for radical chain reactions (e.g. peroxides). Reacts with nitric acid. Risk of spontaneous polymerization in the presence of oxidizing agents.
Hazardous reactions in presence of mentioned substances to avoid.
The product is stabilized against spontaneous polymerization prior to despatch. The product is stable if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

#### Acute oral toxicity

Experimental/calculated data:
LD50rat (oral): 3,150 mg/kg (BASF-Test)

#### Acute inhalation toxicity

LC50 rat (by inhalation): 10.3 mg/l 4 h (OECD Guideline 403)
The vapour was tested.

#### Acute dermal toxicity

LD50 rabbit (dermal): 2,000 - 3,024 mg/kg (other)

#### Assessment of acute toxicity

Of moderate toxicity after short-term inhalation. Of low toxicity after single ingestion. Of low toxicity after short-term skin contact.

**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

### Irritation

Assessment of irritating effects:
Eye contact causes irritation. Skin contact causes irritation.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Irritant. (BASF-Test)

Serious eye damage/irritation rabbit: Irritant. (other)

### Respiratory/Skin sensitization

Assessment of sensitization:
Sensitization after skin contact possible.

Experimental/calculated data:
Mouse Local Lymph Node Assay (LLNA) mouse: skin sensitizing (OECD Guideline 429)

### Germ cell mutagenicity

Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in a test with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

### Carcinogenicity

Assessment of carcinogenicity:
In long-term animal studies in which the substance was given by inhalation, a carcinogenic effect was not observed. The substance showed no carcinogenic activity in animals after chronic administration to the skin. IARC Group 3 (not classifiable as to human carcinogenicity).

### Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

### Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies.

### Specific target organ toxicity (single exposure)

Causes temporary irritation of the respiratory tract.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation. The substance may cause damage to the olfactory epithelium after repeated inhalation.

### Aspiration hazard

not applicable

## 12. Ecological Information

### Ecotoxicity

Toxicity to fish:
LC50 (96 h) 2.1 mg/l, Cyprinodon variegatus (OECD Guideline 203, Flow through.)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic invertebrates:
EC50 (48 h) 8.2 mg/l, Daphnia magna (OECD Guideline 202, part 1, Flow through.)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic plants:
EC50 (96 h) 2.65 mg/l, Selenastrum capricornutum (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

Microorganisms/Effect on activated sludge:
EC0 (3 d) > 150 mg/l, activated sludge, industrial (other, aerobic)
Nominal concentration.

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.136 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Assessment of terrestrial toxicity:
No effects at the highest test concentration.
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Soil living organisms:
EC50 (28 d) > 1,000 mg/kg, soil dwelling microorganisms (OECD Guideline 217)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The details of the toxic effect relate to the nominal concentration.

Terrestrial plants:
No data available.

Other terrestrial non-mammals:
No data available.

### Mobility

Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

### Persistence and degradability

Elimination information:
80 - 90 % TIC of the ThIC (28 d) (ISO 14593) (aerobic, activated sludge, domestic)

Assessment of stability in water:
In contact with water the substance will hydrolyse slowly.

Information on Stability in Water (Hydrolysis):
$t_{1/2}$ 1,100 d (25 °C, pH value 7), (OECD Guideline 111, pH 7)
In contact with water the substance will hydrolyse slowly.

### Bioaccumulation potential

Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Bioaccumulation potential:
Bioconcentration factor: 17.3 (calculated)
Accumulation in organisms is not to be expected.

### Additional information

Other ecotoxicological advice:
Do not release untreated into natural waters. Acutely toxic for aquatic organisms.

## 13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Uncleaned empties should be disposed of in the same manner as the contents.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| UN number or ID number: | UN 2348 |
| UN proper shipping name: | BUTYL ACRYLATES, STABILIZED |
| Transport hazard class(es): | 3 |
| Packing group: | III |
| Environmental hazards: | no |
| Special precautions for user: | None known |

**Sea transport**
IMDG

| | |
|---|---|
| UN number or ID number: | UN 2348 |
| UN proper shipping name: | BUTYL ACRYLATES, STABILIZED |
| Transport hazard class(es): | 3 |
| Packing group: | III |
| Environmental hazards: | no<br>Marine pollutant: NO |
| Special precautions for user: | EmS: F-E; S-D |

**Air transport**
IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 2348 |
| UN proper shipping name: | BUTYL ACRYLATES, STABILIZED |
| Transport hazard class(es): | 3 |
| Packing group: | III |
| Environmental hazards: | No Mark as dangerous for the environment is needed |
| Special precautions for user: | None known |

### Maritime transport in bulk according to IMO instruments

| | |
|---|---|
| Regulation: | IBC-Code |
| Product name: | Butyl acrylate (all isomers) |
| Pollution category: | Y |
| Ship Type: | 3 |

| | |
|---|---|
| ERG-number: | 129P |

See Section 15 for domestic transport regulations

## 15. Regulatory Information

Japanese Fire Service Law: 4th group, Petroleum Group 2, Not water soluble liquid

n-butyl acrylate
Industrial Safety and Health Law (Japan)
MSDS (JP)

n-butyl acrylate
Industrial Safety and Health Law (Japan)
ISHL, Labeling Requirements (JP)

Industrial Safety and Health Law (Japan)

ISHL, Dangerous Substances, Flammables (JP)
listed

Skin and Eye Damaging or Absorbing Substances for PPE Requirement (ISHL Ministerial Ordinance Art. 594-2, MHLW Labour Standard Bureau Circular No. 0704-1), as amended (Japan)
n-butyl acrylate

| Chemical name or elemental component | PRTR (from April 1st, 2023) | | |
|---|---|---|---|
| | Consistency (%) | Classification, Control No. | Cabinet order name |
| n-butyl acrylate | 100 | Class 1, 7 | Butyl acrylate |

Law Relating to the Prevention of Marine Pollution and Maritime Disaster (Japan)
Marine Pollution Prevention Law (JP)
Modulus: 10
Category: Y
Noxious liquid substances which, if discharged into the sea from tank cleaning or deballasting operations, are deemed to present a hazard to either marine resources or human health or cause harm to amenities or other legitimate uses of the sea and therefore justify a limitation on the quality and quantity of the discharge into the marine environment.

**Other regulations**

## 16. Other Information

Safe Handling and Storage aspects are covered in a brochure which is available on request.

Compliant with JIS Z 7252/7253 :2019] Required items that are not described in this SDS indicate that there is no information

Some of the SDS and risk assessment required substances under Industrial Safety and Health Law are indicated with concentration range in SDS for trade secret protection.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.